Jflex, Regex, and Twitter feeds



Twitter Dataset

https://docs.google.com/spreadsheets/d/1CpT3IAhyskHlRmVFLrrHpcOfV57jq_p70D_b3u7EdxI/edit#gid=2093523028

TweetID	Weekday	Hour	Day	Lang	IsReshare	Reach	Retweet	Likes	Klout	Sen	text	LocationID
tw-68271287333	Thursday	17	31	en	FALSE	44	0	0	35	0	We are hiring: Senior Software Engineer - Proto http://www.reqcloud.com/jobs/719865/?k=0LaPxXuFwczs1e32Z	3
tw-68271304535	Thursday	17	31	en	TRUE	1810	5	0	53	2	RT @CodeMineStatus: This is true Amazon Web Services https://aws.amazon.com/ #php #html #html5 #css #w	3
tw-68271321937	Thursday	17	31	en	FALSE	282	0	0	47	0	Devops Engineer Aws Ansible Cassandra Mysql Ubuntu Ruby On Rails Jobs in Austin TX #Austin #TX #jobs #jo	3
tw-68271343696	Thursday	17	31	en	FALSE	2087	4	0	53	0	Happy New Year to all those AWS instances of ours!	3
tw-68271404819	Thursday	17	31	en	FALSE	953	0	0	47	0	Amazon is hiring! #Sr. #International Tax Manager - AWS in #Seattle apply now! #jobs http://neuvoo.com/job.php	3
tw-68271458304	Thursday	17	31	en	FALSE	113	0	0	31	-1	#AWS bc of per-region limits test/prod should be in isolated regions. else dev could impact prod #lambda &	3
tw-68271552643	Thursday	17	31	en	FALSE	113	0	0	31	-2	Isolating dev/prod #aws envs like that is difficult bc of non-ubiquitous service availability. Isn't really a good way a	3
tw-68271720904	Thursday	17	31	en	FALSE	113	0	0	31	-1	Isolation at account level would work. #aws alludes to this on pg 17 but this makes identity management difficult	3
tw-68271796796	Thursday	17	31	en	FALSE	4893	1	0	49	0	Red line vs. green line: Reporting on the most vital Reserved Instance metrics: http://blog.cloudability.com/red-lir	3
tw-68271804808	Thursday	17	31	en	TRUE	2868	6	0	51	2	RT @HPE_BigData: We developed a test drive on AWS. Why is it worth trying? @waltermaguire explains: http://	
tw-68271902295	Thursday	17	31	en	TRUE	86	51	0	16	2	RT @awscloud: AWS CloudFormation now supports AWS WAF & AWS Directory Service for MS Active Dir	3
tw-68271917711	Thursday	17	31	en	FALSE	82	0	0	23	0	AWS Podcasts Banter Predicsis Peak Urban Massage Contentful Babbel and Intel IoT https://aws.amazon.com/	
tw-68271973333	Thursday	17	31	en	FALSE	368	0	0	43	1	Mighty Big Data Set: Month in Review - Lots for big data enthusiasts in December on the AWS Big Data Blog. Ta	3
tw-68271985785	Thursday	17	31	en	FALSE	113	0	0	31	-1	Limited info out there on #aws dev/prod isolation topic so I guess those are the only options: region OR account.	3
tw-68272292567	Thursday	17	31	en	FALSE	8	0	0	21	3	Great read on AWS VPC for startups. #AWSCloud #COMMUNICATE_WELL https://medium.com/aws-activate-s	3
tw-68272309027	Thursday	17	31	en	FALSE	2214	0	0	0	0	Is there a cheaper cloud provider that offers everything AWS does? VPCs ELBs SGs RDS elasticache etc?	3
tw-68272403815	Thursday	17	31	en	FALSE	387	0	0	56	0	Mighty Batman Set: Month in Review - Lots for Batman enthusiasts in December on the AWS Batman Blog. Take	3
tw-68272552526	Thursday	17	31	en	TRUE	149	3	0	31	0	RT @jeffbarr: Some new #AWS Podcasts Banter Predicsis Peak Urban Massage Contentful Babbel and Intel Io	
tw-68272556296	Thursday	17	31	en	TRUE	356	2	0	47	2	RT @Capgemini: How can your business benefit from catalog-based IT services by #Cloud providers like #AWS	3
tw-68272632046	Thursday	17	31	en	TRUE	201	3	0	40	0	RT @jpaulreed: Happy New Year to all those AWS instances of ours!	3
tw-68272675355	Thursday	17	31	en	FALSE	6741	2	0	49	1.3	Learn how to extend #cloud security to #AWS to ensure your organization's data is secure http://info.palerra.com	3
tw-68272833827	Thursday	18	31	en	FALSE	3774	0	0	45	0	"Microsoft Azure vs Amazon AWS Public Cloud Services https://blog.znetlive.com/microsoft-azure-vs-amazon-av	
tw-68272896438	Thursday	18	31	en	FALSE	932	2	0	0	4	How awesome it was to be in #engineering at @Battlefy -- psst they are #hiring! http://blog.battlefy.com/how-we-	3
tw-68272962881	Thursday	18	31	en	FALSE	1768	0	0	46	0	#AWS #/ Cloud Specialist (IT Infrastructure / Architecture) http://neuvoo.co.uk/job.php?id=p8zbummukb&source=	3

Example data

tw-682713045357998080,Thursday,17,31,en,TRUE,1810,5,0,53,2,RT @CodeMineStatus: This is true Amazon Web Services https://aws.amazon.com/ #php #html #html5 #css #webdesign #seo #java #javascript

tw-682713219375476736,Thursday,17,31,en,FALSE,282,0,0,47,0,Devops Engineer Aws Ansible Cassandra Mysql Ubuntu Ruby On Rails Jobs in Austin TX #Austin #TX #jobs #jobsearch https://www.jobfindly.com/devops-engineer-aws-ansible-cassandra-mysql-ubuntu-ruby-on-rails-jobs-austin-tx.html,3741,tw-4624808414

Chosen patterns

Tweet id example: “tw-682712873332805633”

- Starts with “tw-” followed by numbers

Weekday: “Thursday”

- Starts with capital letter and ends with “day”, only has letters

Hashtag : “#love:

- Only letters and numbers are allowed to be part of a hashtag. If a character other than these follows the leading # and a letter or number, the hashtag will be cut off at this point.

Retweet @ (in the text): “RT @BillyBob:”

- Starts with “RT @” and ends with a “:”, username can include characters, underscore, numbers

Also Normal @: “@BobBilly”

Link (in the text): “https://bob.com”

- starts with “https://” and has a “.com” and ends with a space, includes dashes, characters, numbers, slashes, percents, equal, question mark, underscore, parentheses, ampersand, period

Pattern definitions

Up_Letter = [A-Z]

Hashtag = "#"

Symbol = [-/%=?_()&.]

Low_Letter = [a-z]

Letter = [a-zA-Z_]

Digit = \d

Lexical rules

Identifier: "tw-"{Digit}{18}

Weekday: {Up_Letter}{Low_Letter}+"day"

Hashtag: {Hashtag}{Letter}+

Retweet/mention: "RT "?"@"{Letter}*":"

Link: ("https"|"http")"://"({Letter}|{Symbol}|{Digit})*

Example output

<tweet identifier:tw-68271304535799808O>

<weekday:Thursday>

<retweet or mention:RT @CodeMineStatus:>

<link:https://aws.amazon.com/>

<hashtag:#php>

<hashtag:#html>

Conclusion

We used regular expressions to scan and parse data from a corpus of social media tweets.

We generate our Scanner using Jflex.

We provide functionality to detect tweet-ids (label), weekdays it was published (label), mentions and retweets (in each tweet), hashtags (in each tweet), and urls (in each tweet).

References

Inspiration for Regex ideas:

https://www.variables.sh/complex-regular-expression-examples/

Dataset:

https://data.world/a-dukey/twitter/workspace/file?filename=Twitter%2Bdata%2Bin%2Bsheets.xlsx